AURA 2026 Data Quality & Exploratory Analysis Report

Dataset description: 286,696 valid observations were parsed from AY_Project_Data_1786812502784.xlsx. The analysis frame contains split date/time fields, repeated metadata/header rows, and missing sentinel values.

Cleaning methodology: 3,121 non-observation rows were excluded. Exact duplicate rows: 0. Duplicate timestamps: 0; conflicting timestamp groups: 0. The -9999 sentinel was preserved as missing, not converted to zero.

Quality statistics: Signal Quality: valid 100.0%, missing 0.0%, mean 11.047207, median 11.0, std 2.128739, outliers 10350; Signal Strength (dBm): valid 100.0%, missing 0.0%, mean -91.90551, median -92.0, std 4.257128, outliers 10350; Sat. View: valid 55.379%, missing 44.621%, mean 8.008919, median 8.0, std 2.041418, outliers 2452; Sat. Used: valid 3.156%, missing 96.844%, mean 7.587488, median 8.0, std 2.051797, outliers 2; Lat. (deg.): valid 100.0%, missing 0.0%, mean 7.298694, median 7.298822, std 0.064684, outliers 16544; Long. (deg.): valid 100.0%, missing 0.0%, mean 5.13411, median 5.134193, std 0.043373, outliers 15467; Altitude (m): valid 99.995%, missing 0.005%, mean 384.407665, median 385.4, std 30.027146, outliers 14892; Accuracy: valid 99.985%, missing 0.015%, mean 2.291839, median 1.1, std 9.257896, outliers 24945; Temp. (deg. C): valid 100.0%, missing 0.0%, mean 26.12947, median 25.98, std 2.476401, outliers 1802; RH (%): valid 100.0%, missing 0.0%, mean 69.139404, median 75.94, std 20.510107, outliers 17185; Pressure (mbar): valid 100.0%, missing 0.0%, mean 969.941554, median 970.15, std 2.199068, outliers 1633; Heat Index (deg. C): valid 34.795%, missing 65.205%, mean 24.743328, median 24.26, std 2.180761, outliers 1139; Refractivity: valid 100.0%, missing 0.0%, mean 348.4803, median 365.63, std 29.857886, outliers 12579; Battery level (%): valid 100.0%, missing 0.0%, mean 57.756369, median 58.0, std 0.43151, outliers 69676.

Temporal analysis: the observation frame runs from 2026-01-03 00:00:26 to 2026-12-09 23:57:23. April versus July means were {"April 2026": {"temperature": 25.1325, "relativeHumidity": 52.0312, "pressure": 968.9413, "heatIndex": 24.3313, "refractivity": 321.8232}, "July 2026": {"temperature": 26.6256, "relativeHumidity": 77.897, "pressure": 970.8459, "heatIndex": 24.668, "refractivity": 363.7233}}. Daily refractivity statistics and the complete daily profile are provided in the analytics assets.

Relationships: temperature Pearson r=-0.001822, Spearman r=0.257563, n=286696; relativeHumidity Pearson r=0.94223, Spearman r=0.704184, n=286696; pressure Pearson r=0.704099, Spearman r=0.559734, n=286696; heatIndex Pearson r=-0.720896, Spearman r=-0.619202, n=99757. Correlations are descriptive associations and do not establish causation.

Chronological split: 202,965 records from February-August 2026 were eligible for training; 74,021 records from September 2026 onward were held out. No holdout rows were used for model fitting, feature fitting, or validation.

Model: Chronological multivariate linear regression targeting refractivity. Validation used the final training month as a chronological validation period. Metrics: MAE=3.193933, RMSE=3.972661, R2=0.910511, bias=1.037282, MAPE=0.883242.

Forecast statistics: monthly refractivity and environmental summaries were September 2026: temperature 26.2418, RH 66.5113, pressure 969.84, refractivity 345.2927; October 2026: temperature 26.2275, RH 66.7922, pressure 969.8656, refractivity 345.6343; November 2026: temperature 26.2478, RH 66.7181, pressure 969.8753, refractivity 345.5977; December 2026: temperature 26.2292, RH 66.7119, pressure 969.8627, refractivity 345.5466. Future environmental predictors use reproducible training-period weekday-by-hour profiles, with the

selected hour and date reflected in each forecast row.
Package coverage: the Excel outputs include analysis/statistics sheets, while the analytics directory contains missingness bars, outlier bars, histograms, correlation charts, daily trend, validation metrics, training/holdout counts, and monthly forecast charts.